

SUNCHEON, Korea, Republic ofWMO: **471740**Lat: **35.0203N**Lon: **127.3694E**Elev: **167**StdP: **99.34**Time Zone: **9.00 (E09)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-7.1	-17.8	0.8	-1.7	-15.7	1.0	-1.5	7.0	0.5	6.4	-0.3	1.6	320	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	32.5	24.9	31.3	24.6	30.0	23.8	26.1	30.3	25.6	29.3	25.1	28.4	2.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.6	27.9	24.6	20.0	27.5	24.1	19.4	27.1	82.2	30.5	79.9	29.4	77.8	28.6	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.4	4.9	-10.8	34.5	1.6	1.2	-11.9	35.4	-12.9	36.1	-13.7	36.8	-14.9	37.7		
(5)			-11.6	26.8	1.4	0.9	-12.6	27.4	-13.4	27.9	-14.1	28.4	-15.1	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	0.0	2.5	6.9	12.0	17.5	21.6	25.3	25.8	21.2	14.9	8.6	2.0
	DBStd	9.35	3.32	3.75	3.36	3.14	2.54	1.86	2.13	2.28	2.62	3.01	3.99	3.42
	HDD10.0	972	310	212	109	16	0	0	0	0	0	2	73	249
	HDD18.3	2519	568	444	356	189	45	1	0	0	5	112	293	507
	CDD10.0	2159	0	2	12	78	233	348	475	489	334	156	31	1
	CDD18.3	664	0	0	0	1	20	99	217	231	89	7	1	0
	CDH23.3	5172	0	0	0	18	271	576	1729	2049	506	24	1	0
	CDH26.7	1497	0	0	0	0	44	103	528	732	90	0	0	0
Wind	WSAvg	1.8	2.3	2.1	2.2	2.1	1.8	1.5	1.4	1.4	1.1	1.3	1.7	2.2
Precipitation	PrecAvg	1431	30	44	64	92	104	170	311	319	154	57	53	33
	PrecMax	1992	84	128	136	228	215	349	659	753	513	200	147	68
	PrecMin	831	4	8	14	22	40	21	27	115	23	3	4	5
	PrecStd	287	16	29	32	48	43	93	136	151	115	43	37	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	16.4	20.9	25.6	29.8	30.9	33.4	34.5	31.0	25.6	21.8	14.1
		MCWB	6.8	13.8	11.7	13.7	17.5	20.7	25.5	25.3	23.7	18.8	16.5	10.2
	2%	DB	9.3	13.8	18.6	23.2	27.8	29.1	32.1	33.1	28.9	23.7	18.8	11.7
		MCWB	5.0	9.2	10.5	13.3	16.4	20.4	25.3	24.9	22.0	17.0	14.2	8.2
	5%	DB	7.3	11.7	16.2	21.2	26.4	27.8	30.9	31.9	27.5	22.5	17.2	9.9
		MCWB	3.2	6.6	9.6	12.6	16.2	20.2	24.9	25.0	21.3	16.4	13.1	6.7
	10%	DB	5.6	9.2	13.9	19.2	24.6	26.6	29.6	30.6	26.2	21.1	15.5	7.9
		MCWB	2.4	5.3	9.3	11.7	16.0	20.0	24.3	24.8	20.6	16.1	11.9	4.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	14.5	14.7	17.0	20.9	24.1	26.9	27.0	24.9	21.6	17.9	12.2
		MCDB	10.5	16.1	16.8	21.2	22.2	26.6	31.3	31.6	28.8	23.0	19.9	13.0
	2%	WB	5.5	11.1	13.0	15.7	19.7	23.2	26.2	26.2	24.0	20.1	16.1	9.3
		MCDB	8.0	12.9	14.8	19.2	22.9	25.7	30.1	30.6	26.6	22.5	17.8	11.3
	5%	WB	4.4	7.7	11.5	14.6	18.4	22.2	25.6	25.7	23.2	18.5	14.1	6.8
		MCDB	6.6	10.2	15.1	17.8	22.6	25.2	29.2	29.7	25.8	20.1	16.3	9.1
	10%	WB	2.7	5.6	9.6	13.5	17.5	21.4	25.1	25.2	22.2	16.6	12.5	5.3
		MCDB	5.3	8.7	13.6	17.1	22.0	24.8	28.4	28.8	24.6	19.6	14.8	7.7
Mean Daily Temperature Range	5% DB	MDBR	9.0	10.2	11.8	12.1	12.6	9.1	6.8	7.7	9.4	11.6	10.9	9.2
		MCDBR	11.7	13.9	15.9	16.5	16.7	12.0	9.2	10.0	10.5	13.0	12.8	12.3
		MCWBR	7.7	8.7	8.7	7.6	6.7	4.7	3.3	3.3	4.2	6.1	7.6	8.4
	5% WB	MCDBR	10.6	10.8	12.0	10.7	11.1	7.8	7.8	8.2	7.4	7.7	9.4	11.2
MCWBR		7.9	8.5	8.4	7.0	5.8	3.9	3.1	3.2	3.5	5.0	6.5	8.5	
Clear-Sky Solar Irradiance	taub	0.422	0.464	0.538	0.559	0.574	0.629	0.591	0.547	0.486	0.454	0.434	0.419	
	taud	2.002	1.936	1.781	1.785	1.795	1.746	1.897	2.007	2.122	2.082	2.050	2.024	
	Ebn at Noon	739	755	734	746	741	699	723	747	774	759	722	712	
	Edh at Noon	137	164	208	217	218	229	195	172	146	140	130	127	
All-Sky Solar Radiation	RadAvg	2.60	3.33	4.37	5.00	5.46	4.79	4.40	4.54	4.06	3.69	2.71	2.33	
	RadStd	0.19	0.28	0.38	0.40	0.58	0.46	0.63	0.60	0.43	0.33	0.28	0.14	